Exponential Growth and Decay

Questions 1–26

Question 1. For each of the equations below find the value of y when $x = 0$:

a) $y = 1 + 2^{-x}$
b) $y = 20 - 2^{0.1x}$
c) $y = 10 + e^{-0.1x}$
d) $y = 5 - e^{0.001x}$
e) $y = 21 + 5e^{-0.5x}$
f) $y = 3 - 5 \times (\frac{1}{2})^{-\frac{1}{4}x}$

Question 2. Find the value of P when $t = 0$ for the following:

a) $P = \frac{100}{1+e^{-\frac{1}{2}t}}$
b) $P = \frac{18}{1+e^{-0.1t}}$

Question 3. A population of bacteria is modelled with the equation

$$P = 250e^{0.15t}$$

where P is the population size and t is the time in hours.

a) State the size of the initial population.
b) Find the size of the population after 20 hours.
c) When does the population reach 1000 bacteria?

Question 4. For each of the models below, find the time t (hours) it takes for an amount of a substance y (grams) to double from its initial amount, giving your answers in exact form.

a) $y = 3 \times 2^t$
b) $y = 4e^{1.5t}$
c) $y = 150 \times 1.6^t$
d) $y = 20 + 4e^{0.1t}$

Question 5. The quantity of radioactive isotope Uranium-235 y grams over time may be modelled by the equation

$$y = 1000 \times 2^{-0.00142t}$$

where t is millions of years.

a) State the initial mass of the Uranium-235

b) Show that the half-life of Uranium-235 is about 704 million years.

Question 6. For each of the models below, find the initial value of y and state what is happening to the value of y as $x \to \infty$

Hence sketch each of the models.

a) $y = 10 + 2e^{-0.3x}$

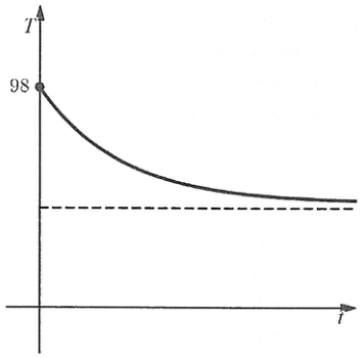
b) $y = -5 + 7e^{-2x}$

c) $y = 1.2 - 0.6e^{-7.6x}$

d) $y = 0.2 + 1.8e^{2.3x}$

Question 7. The graph below shows the temperature of a cup of coffee over time.
The room temperature is 22°C.
The temperature, T°C, after t minutes may be modelled by the equation

$$T = A + Be^{-\frac{1}{5}t}$$



a) State the initial temperature of the coffee.
b) Write down the equation of the horizontal asymptote.
c) Hence find the values of A and B
d) Suggest how the model may be refined for a cup of coffee that is initially 100°C.
e) Suggest values of A and B for a cup of coffee cooling at the same rate but from initial temperature 97°C in a room at 18°C.

Question 8. The height of a plant is given by the equation

$$h = \frac{50}{1 + 4e^{-\frac{1}{5}t}}$$

where h is the height in cm and t is the number of weeks that have passed.
a) According to this model state the initial height of the plant.
b) What height does this model predict in the long term for this plant?

Question 9. The height of a child was recorded for the first seven years of their life. A model suggested for the child's height, h cm, at x years old is given by

$$h = 130 - 80 \times 1.45^{-x}$$

where $0 \leq x \leq 7$. Using the model:

a) State the height of a newborn baby.
b) Find the height of a child after 18 months, giving your answer to the nearest millimetre.
c) Estimate the age of the child in months (to the nearest month) when their height is 1 m.
d) Explain why this model might be unsuitable for predicting the child's height as a teenager.

Question 10. An investment of £1200 is made at 5% compound interest per annum, with interest added to the balance only at the end of each complete year.

a) Create an equation to model the value of the investment over time.
b) Sketch the curve for your model, including labelling any points of intersections with axes.
c) Find the value of the investment after 4 years giving your answer to the nearest penny.
d) Find when the investment reaches more than £2000.

Question 11. A fine for speeding is initially £A. Each week it is not paid, the fees increase exponentially. After 1 week the fine becomes £150 and after 2 weeks it becomes £225.

a) Calculate the initial fine, £A

b) Find a model which for this data, giving your answer in the form

$$C = A \times r^t$$

where C is the cost, t is the time (in weeks) and A and r are constant values to be found.

c) Use your model to find the cost of the fine in week 4.

Question 12. The number of people using a social media platform is declining exponentially since its peak of popularity (at $t = 0$).
At 3 years and 6 years it had 180 and 144 million active users respectively.

a) Find the number of active users at its peak.

b) Hence find a model which fits the data, giving your answer in the form

$$N = A \times r^{\frac{1}{3}t}$$

where N is the number of active users, in millions, t is the time, in years, and A and r are constant values to be found.

c) Use your model to predict the number of users after 10 years has passed.

Question 13. A population of insects is known to be growing exponentially. The population 2 and 4 months into the data collection is 1440 and 3240 insects respectively.
A model for the data may be given by

$$P = A \times r^t$$

where P is population, t is time (in months) and A and r are constant values to be found.

a) Find the values of A and r in the model.
b) Interpret the value of the constants A and r in context.

Question 14. The amount of stock in a warehouse of a particular product P, is known to be 12 000 units. After 5 months this is expected to have reduced by 9500.

a) Find a model of the form

$$P = ae^{kt}$$

where t is the number of months passed, where a and k are constant values to be found.
b) How much stock is available after 2 months?
c) The company orders a delivery from the manufacturers when the stock drops to 500 units. After how many months should they make this order?
d) What does this model predict in the long term for the stock of this product? Comment on its reliability.

Question 15. The amount, C milligrams, of a particular drug in a person's body after t hours may be given by a function of the form

$$C = Ae^{-kt}$$

a) Given that the initial dose taken is 200mg and in 4 hours the amount of drug has halved once, find the values of A and k giving your answer in exact form.
b) Hence, find the number of hours it takes for the drug to reduce to 50 mg in a person's body.

Question 16. The number P of restaurants in a particular fast food chain in their nth year may be modelled by the function

$$P = Ae^{kn}$$

In the first year ($n = 1$), there were 1000 restaurants and in the second year there were 1400.
a) Show that $e^k = 1.4$
b) Find the exact value of k and A.
c) How many restaurants does the model predict in the 4th year?
d) How many whole years is it before there are 100 000 restaurants?
e) Explain why the model is invalid in the long-term.

Question 17. A substance p is known to decay exponentially according to the model

$$p = p_0 e^{-kt}$$

where t is time in years, p_0 is the initial quantity and k is an unknown constant.
Find the amount of time needed for the quantity to have reduced by 99%, giving your answer in the form $\frac{1}{k} \ln A$ where A is a positive integer.

Question 18. The amount of a drug y mg in the blood stream is modelled by the equation

$$y = Ae^{-kt}$$

where t is the time in hours since a tablet was taken. After 2 hours there was 500mg in the blood stream, and after 7 hours there was 88mg in the blood stream.

a) Find the exact value of k
b) Find the value of A to 1 decimal place.

Question 19. The populations of two types of fish in a lake may be modelled with the following equations:
Fish A:

$$P_A = ae^{0.08t}$$

Fish B:

$$P_B = be^{0.12t}$$

where P_A and P_B are the respective populations of fish A and fish B and t is the number of months passed since the start of the year.
The total number of fish in the lake at the start of the year is 85, then after 6 months passes the total population is 153.

a) Find the values of a and b giving your answers to 1 decimal place.
b) Interpret what your values for a and b mean in context.
c) Find the number of months passed when the population of fish B first exceeds the population of fish A.
d) Explain why the models are invalid in the long-term.

Question 20. A beaker of liquid is cooling. Its initial temperature is 65°C.
After 3 minutes its temperature is 35°C.
The temperature is modelled using the equation

$$T = 17 + ae^{-kt}$$

where T is the temperature in °C, t is the time in minutes and a and k are constants.

a) Find the exact values of a and k.
b) What is the temperature of the liquid after 5 minutes?
c) What is the temperature of the liquid in the long term?
d) State how the model could be modified if the room temperature is reduced.

The temperature of a second liquid, y°C, cooling from 80°C at the same point in time in the same room is modelled by the equation

$$y = A + be^{-1.5kt}$$

where k has the same value from before.

e) Find the time at which both liquids have the same temperature.

Question 21. Prerequisite knowledge for Q21-Q26: Differentiation of the form ae^{kx} For each of the equations calculate the rate of change in y when $x = 2$:

a) $y = 20 + 2e^{-0.3x}$
b) $y = 4(5 + e^{0.2x})$
c) $y = 70 - 60e^{-0.8x}$

Question 22. The population of red squirrels in the UK can be modelled by the equation

$$P = 3500e^{-\frac{1}{225}\ln(\frac{700}{57})t}$$

where P is their population (in thousands) and t is the number of years since 1800.
Show that in 2000, according to the model, the amount of squirrels is decreasing at a rate of approximately 4.2 thousand per year.

Question 23. A volume V (ml) of liquid being poured from a bottle may be modelled by the equation

$$V = 330(1 - e^{-0.1t})$$

after t seconds.

a) State the theoretical maximum amount of liquid the bottle is holding, according to the model.

b) Find the rate at which the liquid is being poured when $t = 5$ seconds.

Question 24. The amount of stock in a shop of two products P may be modelled by the following equations where t is the number of months since the start of the year.

Product A: $P = 100(1 + e^{-kt})$

Product B: $P = 400e^{-kt}$

where $t \geq 0$ and k is a positive constant.

a) Sketch the two models on the same axes, include labelling points where the curves meet the axes and any asymptotes.

b) State the amount of each product in the long term according to the model.

c) Explain the limitations of the model.

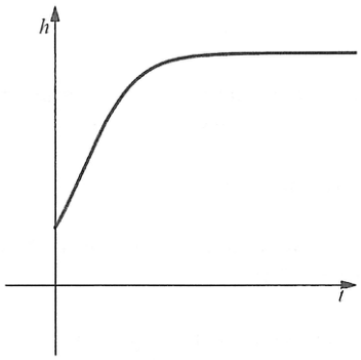
d) Show that the rate of decrease of product B is always 4 times that of product A.

e) Find the exact value of k given there is the same amount of each product after two months have passed.

Question 25. The height of a tree is given by the equation

$$h = \frac{18e^{\frac{1}{2}t}}{e^{\frac{1}{2}t} + 3}$$

where h is the height in metres and t is the number of years that have passed since it is planted. The graph of height against time is shown below.



a) According to this model state the initial height of the tree.
b) Show that

$$h = \frac{18}{1 + 3e^{-\frac{1}{2}t}}$$

c) What height does this model predict in the long term for this tree?

Question 25 (continued). A second tree was planted at the same time, and it may be modelled with equation

$$h = a - 14e^{-\frac{1}{2}t}$$

where a is an unknown constant.
d) Find the value of a given this tree has a height of 2 m when planted.
e) Find the rate of growth of the second tree at 5 years.
f) State which of the two trees is expected to be taller.

Question 26. Two models of mass m kg over an exponentially decaying substance over time t days are given by

$$m = 100e^{-0.05t}$$

and

$$m = 10 + 90e^{-kt}$$

where k is a positive constant.
Find the ranges of values of k for which the second model is initially decaying at a faster rate than the first.